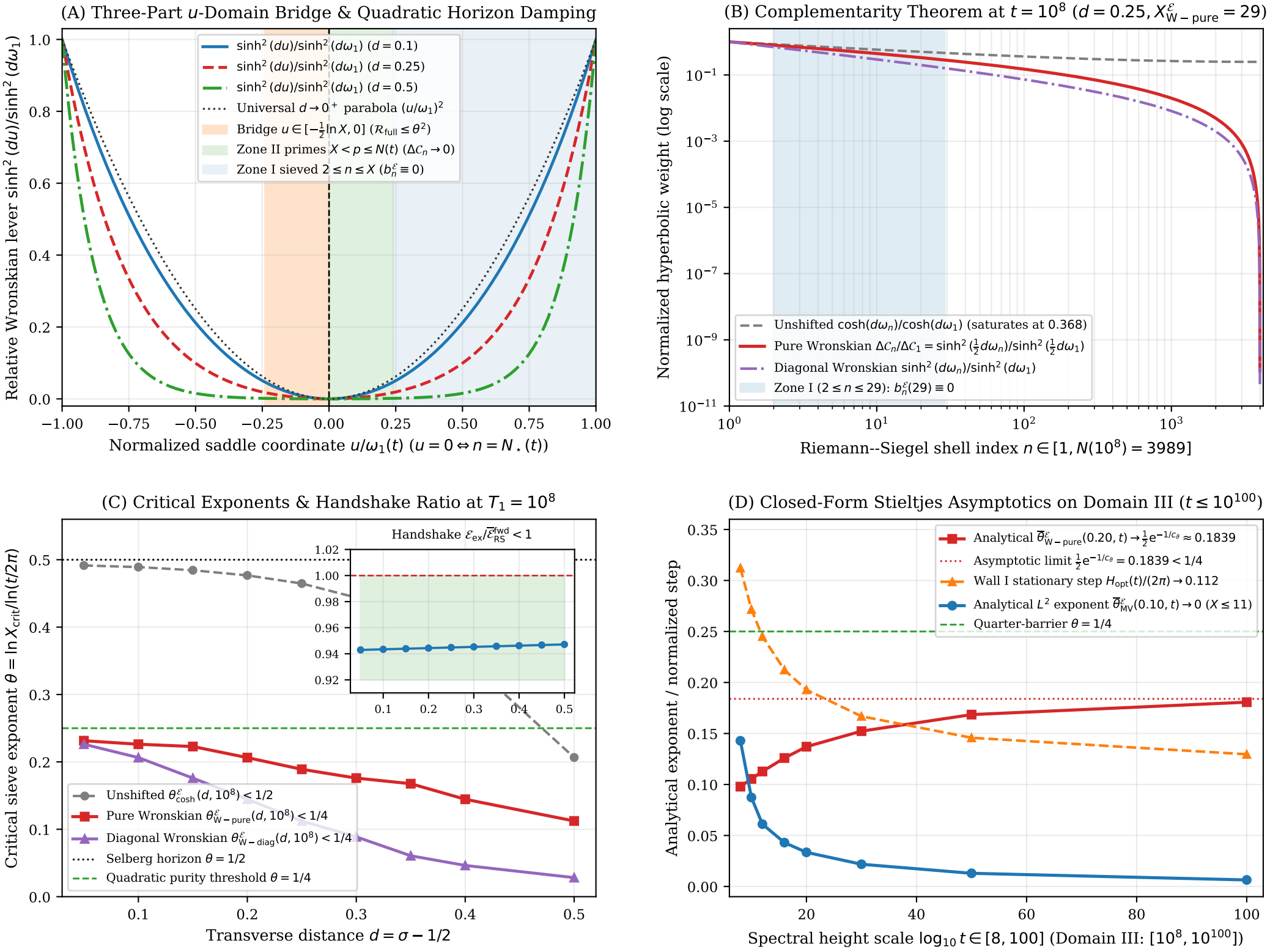

(A) Three-Part u-Domain Bridge & Quadratic Horizon Damping
Relative Wronskian lever $\sinh^2(du)/\sinh^2(d\omega_1)$
Normalized saddle coordinate $u/\omega_1(t)$ ($u=0 \Leftrightarrow n=N_\star(t)$)
$\sinh^2(du)/\sinh^2(d\omega_1)$ ($d=0.1$)
$\sinh^2(du)/\sinh^2(d\omega_1)$ ($d=0.25$)
$\sinh^2(du)/\sinh^2(d\omega_1)$ ($d=0.5$)
Universal $d\to 0^+$ parabola $(u/\omega_1)^2$
Bridge $u\in[-\frac{1}{2}\ln X, 0]$ ($\mathcal{R}_{\text{full}}\le\theta^2$)
Zone II primes $X<p\le N(t)$ ($\Delta C_n\to 0$)
Zone I sieved $2\le n\le X$ ($b_n^{\mathcal{E}}\equiv 0$)
(B) Complementarity Theorem at $t=10^8$ ($d=0.25$, $X^{\mathcal{E}}_{\text{W-pure}}=29$)
Normalized hyperbolic weight (log scale)
Riemann--Siegel shell index $n\in[1, N(10^8)=3989]$
Unshifted $\cosh(d\omega_n)/\cosh(d\omega_1)$ (saturates at 0.368)
Pure Wronskian $\Delta C_n/\Delta C_1=\sinh^2(\frac{1}{2}d\omega_n)/\sinh^2(\frac{1}{2}d\omega_1)$
Diagonal Wronskian $\sinh^2(d\omega_n)/\sinh^2(d\omega_1)$
Zone I ($2\le n\le 29$): $b_n^{\mathcal{E}}(29)\equiv 0$
(C) Critical Exponents & Handshake Ratio at $T_1=10^8$
Critical sieve exponent $\theta=\ln X_{\text{crit}}/\ln(t/2\pi)$
Transverse distance $d=\sigma-1/2$
Handshake $\mathcal{E}_{\text{ex}}/\overline{\mathcal{E}}^{\text{fwd}}_{\text{RS}}<1$
Unshifted $\theta^{\mathcal{E}}_{\cosh}(d,10^8)<1/2$
Pure Wronskian $\theta^{\mathcal{E}}_{\text{W-pure}}(d,10^8)<1/4$
Diagonal Wronskian $\theta^{\mathcal{E}}_{\text{W-diag}}(d,10^8)<1/4$
Selberg horizon $\theta=1/2$
Quadratic purity threshold $\theta=1/4$
(D) Closed-Form Stieltjes Asymptotics on Domain III ($t\le 10^{100}$)
Analytical exponent / normalized step
Spectral height scale $\log_{10} t\in[8,100]$ (Domain III: $[10^8, 10^{100}]$)
Analytical $\overline{\theta}^{\mathcal{E}}_{\text{W-pure}}(0.20,t)\to\frac{1}{2}e^{-1/c_\theta}\approx 0.1839$
Asymptotic limit $\frac{1}{2}e^{-1/c_\theta}=0.1839<1/4$
Wall I stationary step $H_{\text{opt}}(t)/(2\pi)\to 0.112$
Analytical L^2 exponent $\overline{\theta}^{\mathcal{E}}_{\text{MV}}(0.10,t)\to 0$ ($X\le 11$)
Quarter-barrier $\theta=1/4$